

”Enhanced Fine-Grained Car Classification Using Focal Loss and Part-Based Voting Models”

Luca Fusaro

Email: luca.fusaro@studenti.unipd.it

Repository: <https://github.com/lucafusaro/FineGrainedCompCars>

Abstract—Car classification tasks, especially fine-grained classification, pose significant challenges due to the large number of visually similar models and the inherent class imbalance in datasets like CompCars. In this work, we build on the foundational CompCars dataset by introducing enhancements to address these challenges, particularly focusing on improving classification accuracy for underrepresented classes. We apply Focal Loss, a technique designed to handle class imbalance by giving more importance to difficult-to-classify samples, significantly improving performance on minority classes. In addition to using full car images for classification, we extend the task by incorporating car part classification (e.g., headlights, taillights, dashboards) and introducing a weighted voting strategy that combines the outputs of part-specific models. Our results show that by applying Focal Loss, we achieve a lowest F1 score of 0.5000, substantially higher than 0.3529 achieved using CrossEntropy for car make classification. Furthermore, we obtained a mean balanced model accuracy of 94.35% demonstrating the effectiveness of our approach. Our enhancements provide more robust solutions to car make-model classification, particularly for underrepresented classes, and offer insights into improving performance through part-based features and weighted voting strategies.

Index Terms—Fine-Grained Classification, CompCars Dataset, Convolutional Neural Network (CNN), Class Imbalance, Focal Loss, Car Parts Classification.

I. INTRODUCTION

Fine-grained car classification, which involves distinguishing between visually similar car makes and models, presents significant challenges due to the large number of categories and class imbalance issues. Leveraging the large-scale **CompCars** dataset, this paper proposes an enhanced method to address these challenges, improving performance for underrepresented classes.

Traditional classification approaches, while effective for majority classes, often fail to handle underrepresented classes effectively. In this paper we use a **Resnet34** architecture, and introduce the use of **Focal Loss** to tackle this issue achieving significant improvements in accuracy, particularly for minority classes.

Furthermore we introduce a **weighted voting strategy** across part-specific models for the car parts classification, improving classification performance.

This paper is structured as follows. In Section II, we review related work on car classification and its challenges. Section III presents the processing pipeline, while Section IV describes the dataset and feature extraction process. Section V outlines the learning framework and methods used in this work. Finally,

we present the results in Section VI and conclude with remarks on future research in Section VII. **The main contributions of this paper are as follows:**

- We apply **Focal Loss** to address the class imbalance problem in the CompCars dataset, significantly improving performance on underrepresented car makes and models.
- In fine-grained car parts classification we implement a **weighted voting strategy** for combining predictions from part-specific models, assigning higher weights to car parts with higher accuracy, resulting in improved overall classification performance.
- We demonstrate that using **ResNet34**, a relatively shallow architecture compared to deeper networks like GoogLeNet, combined with Focal Loss and the voting strategy, achieves competitive results with fewer parameters.

II. RELATED WORK

Several studies have addressed the challenges of car classification, leveraging both full-car images and car part images for fine-grained classification tasks. The CompCars dataset, introduced by Yang et al. [1], has been foundational in this field, offering a large-scale dataset with a rich hierarchy of car makes and models. In their approach, the authors employed a Convolutional Neural Network (CNN) architecture, specifically **Overfeat**, to classify car images into car models. However, one limitation noted in their work was the **class imbalance** across car models, particularly for underrepresented models, which was not explicitly addressed.

In the second version of their paper, Yang et al. expanded on their results by comparing different CNN architectures. Their best-performing model, **GoogLeNet**, achieved a **Top-1 accuracy of 91.2%** and a **Top-5 accuracy of 98.1%**, but the class imbalance issue remained unaddressed.

To address this challenge, we implement Focal Loss that was introduced by Lin et al. [2], which adjusts the standard cross-entropy loss by adding a modulating factor that down-weights the loss for well-classified examples. This approach has shown promise in dense object detection and has been adapted in our work to mitigate the impact of class imbalance on car make and model classification. Our work employs **ResNet34**, a 34-layer architecture that, while less deep than GoogLeNet, leverages skip connections to ensure efficient training of deeper networks. The use of the Focal Loss specifically addresses the class imbalance problem, particularly improving performance

on minority classes. Additionally, while the original paper employed a voting strategy to combine predictions from individual car part models, our work introduces a weighted voting strategy, improving classification performance by weighting predictions based on part-specific model accuracy.

III. PROCESSING PIPELINE

In this section, we describe the approach employed for the fine-grained classification of cars using the **CompCars** dataset. The primary architecture used was a **ResNet34** model, initialized with **pretrained weights from ImageNet** to leverage the transfer learning benefits, particularly due to the relatively limited size of the dataset. To ensure reproducibility, seeds were manually set for both CPU and GPU operations. We tackled three classification tasks:

- 1) **Car Make Classification:** Initially, we focused on car make classification, which involved **75 different classes**. For this task, a **cross entropy loss** function was employed. To optimize the convergence during training, different **learning rate schedulers** were explored. Given the significant class imbalance in the dataset, particularly for underrepresented car makes, we experimented with **Focal Loss** to address this imbalance. Various parameter combinations for Focal Loss were tested. Additionally, data augmentation techniques were applied to those classes with fewer examples to further combat the imbalance.
- 2) **Car Make-Model Classification:** For the car make-model classification task, the pretrained ResNet34 model (from the car make classification task) was further **fine-tuned** for both levels of classification (make and make-model). Here, both cross entropy loss and focal loss are tested for make classification. The fine-tuning process helped transfer the learned features from car make classification, improving the model's performance on this task. The diagram in Figure 1 shows the main steps of this pipeline.
- 3) **Car Parts Classification:** In a separate task, we worked on fine-grained classification using car part images. The same ResNet34 architecture was employed, focal loss is used for make classification task.

During the **test phase**, a comprehensive evaluation was carried out using several performance metrics to assess the model's effectiveness. These metrics include accuracy and balanced accuracy scores for make and model, F1 score, confusion matrices. Furthermore, a micro-averaged ROC curve was plotted to assess the model's overall performance across all classes and plots of training and validation loss/accuracy curves to check convergence are included. It is important to note that model accuracy is calculated only when the make prediction is correct, and only for makes that have more than one model.

IV. DATA AND FEATURES

In this project, the data consist of images from the *CompCars* dataset, which provides a rich and diverse collection of

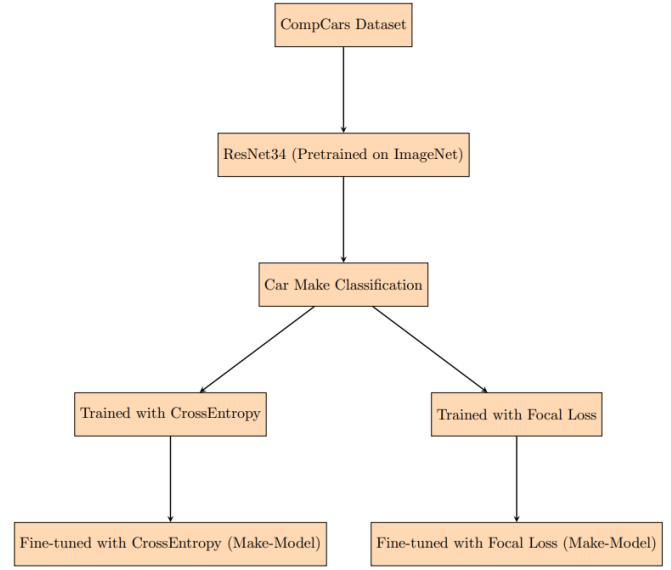


Fig. 1: Training Pipeline for Car Make and Fine-Grained Classification with full car images

car images with detailed annotations. We used the train/test splitting suggested by the authors: the dataset includes **75 car make classes**, with each class potentially containing **1 to 30 car models**. Additionally, for the *car parts classification* task, a different set of images from the *CompCars* dataset was used, specifically focusing on vehicle parts. This set contains images highlighting **4 interior parts** (i.e. console, steering wheel, dashboard, and gear lever) and **4 exterior parts** (i.e. headlight, taillight, fog light, and air intake) of the vehicle.

A. Setup and Input Data Format

The dataset for full car images provides specific coordinates for cropping the images. These coordinates were crucial for focusing the model on the car itself, excluding extraneous parts of the image such as background elements significantly improving model performance by enabling the model to learn relevant *features* such as car shapes, logos, while ignoring irrelevant backgrounds.

B. Preprocessing

After cropping the images, a range of *preprocessing techniques* were applied to enhance the training data:

- **Resizing:** All input images were processed into fixed sizes to align with the input requirements of the **ResNet34** architecture used in this work.
- **Normalization:** All images were normalized with the standard normalization values for RGB images pretrained on the ImageNet to ensure consistent input scales across the dataset.
- **Data Augmentation:** Various augmentation techniques were applied to increase the diversity of training samples. For the *car make* and *car make-model* classification tasks, transformations such as *RandomRotation*,

RandomHorizontalFlip, and ColorJitter were used. These augmentations helped the model generalize better by introducing variability in image orientation, color, and lighting conditions, effectively preventing the model from overfitting to specific visual patterns.

Additionally, when experimenting with *Focal Loss* to address the class imbalance, further augmentation was applied specifically to car make classes with fewer than 50 samples. This additional augmentation included transformations like RandomHorizontalFlip, RandomRotation, and RandomPerspective set to introduce slight distortions and rotations to help the model learn more robust features for the underrepresented classes.

For the *car parts classification* task, the preprocessing pipeline was similar. Similar augmentations were applied to ensure the model could handle variations in positioning, here color jittering is excluded because we experimented that doesn't improve the learning.

C. Feature Extraction

The preprocessed images were then passed through the **ResNet34** architecture, pretrained on the *ImageNet* dataset. This network served as a feature extractor, leveraging the rich feature hierarchies learned during pretraining. The images were fed into the network, which automatically extracted relevant features through its convolutional layers.

The model, pretrained on ImageNet, was initially able to detect general shapes and textures, which were then fine-tuned for the specific task of car make classification. The network's deeper layers were progressively trained to recognize finer details specific to the *CompCars* dataset (e.g., car logos, grilles, or wheel shapes).

D. Dataset Splitting

The dataset used the official train/test split provided by the original *CompCars* paper, ensuring consistency with prior work. Additionally, the training set was further split into **80% for training** and **20% for validation** to assess model performance during training and fine-tune the hyperparameters accordingly. This splitting strategy ensured that the model could be evaluated on unseen validation data before final testing, providing a reliable estimate of its performance on generalization.

For the **car parts classification task**, we used the official train/test split, allowing for comparison with established baselines from the literature. Here, we trained and tested the model separately for each of the **8 car part types**. The data were split in a similar fashion (80/20 for training and validation) for each part. After training individual part-specific models, a **weighted voting strategy** was implemented to aggregate predictions from all part-specific models and provide a final prediction for the test images.

The dataset for the voting strategy was constructed by loading and organizing images by car make and model, associating each part with its respective car. Each car in the

test set had multiple part images (if available), including, for example, images of its headlights, taillights, and dashboard. For cars with missing parts (i.e., when a particular part image was not available), we used a placeholder tensor to maintain consistency in input dimensions. The weighted voting strategy was based on the **mean balanced model accuracy** for each car part type, assigning higher weights to parts with higher balanced accuracies. This allowed the final prediction to rely more heavily on part-specific models that demonstrated higher confidence and accuracy.

V. LEARNING FRAMEWORK

In this section, we describe the learning framework implemented for the tasks of car make classification, car make-model classification, and car parts classification using the *CompCars* dataset.

A. Model Architecture and Training Setup

The core architecture used for all tasks was the **ResNet34** model. The final fully connected layer of the network was replaced to fit the specific number of classes for each task: car make classification, car make-model classification, and car parts classification.

For each task, we applied standard data preprocessing techniques, including resizing images to 224×224 , followed by data augmentation (random rotation, horizontal flipping, and color jittering) to ensure that the model generalizes well and avoids overfitting. The images were also normalized using the standard ImageNet normalization values ($\text{mean} = [0.485, 0.456, 0.406]$ and $\text{std} = [0.229, 0.224, 0.225]$).

The dataset was split into training, validation, and testing sets, using the train/test split provided by the original dataset paper. Additionally, 20% of the training data was set aside for validation.

B. Learning Rate Schedulers and Optimization

In all cases, the training process was optimized using the **Adam** optimizer, with an initial learning rate of 0.001. Various **learning rate schedulers** were tested to improve convergence during training. In the end Cosine Annealing Scheduler is chosen.

C. Loss Functions

Cross-Entropy Loss was the primary loss function used for the car make and car make-model classification tasks. However, due to the significant class imbalance observed in the dataset, **Focal Loss** was introduced in the car make classification task to address this imbalance.

1) *Focal Loss Configuration*: Focal Loss was particularly useful for underrepresented car makes, where standard cross-entropy loss failed to provide robust performance. Several configurations of Focal Loss were experimented with, where we varied the parameters γ and α . The loss function was defined as:

$$\text{Focal Loss} = \alpha_t \cdot (1 - p_t)^\gamma \cdot \text{CrossEntropyLoss}$$

where p_t is the predicted probability of the true class, and α_t is the weight assigned to the class. α values were computed as an interpolation between the square root of the inverse class frequency and a fixed value of 0.5, controlled by η ; the square root was employed to smooth the distribution of values avoiding too large values.

$$\alpha_{\text{interp}} = \eta \cdot \alpha + (1 - \eta) \cdot 0.5$$

Three configurations of Focal Loss were tested:

- **Configuration 1:** $\gamma = 1.0$, $\eta = 0.2$
- **Configuration 2:** $\gamma = 1.5$, $\eta = 0.4$
- **Configuration 3:** $\gamma = 1.5$, $\eta = 0.6$

The second configuration provided the best balance, significantly improving the classification performance for the minority classes. With Focal Loss, performance on minority classes was improved, although the performance on majority classes could slightly diminish due to the increased focus on underrepresented classes.

Focal Loss was not used for model classification because the class imbalance at that level is less severe than for car make classification.

D. Fine-Tuning Strategy for Car Make-Model Classification

For the **car make-model classification** task, we fine-tuned the model by first training the network for car make classification. The pretrained model from this task was then used as a starting point for the make-model classification task. Specifically, ResNet34 architecture was modified implementing two separate heads: one for predicting the car make and the other for predicting the model based on the predicted make. This strategy allowed the model to leverage shared features between the car make and model, improving the overall performance.

E. Evaluation Metrics

During the test phase, several metrics were employed to assess the performance of the models:

- **Accuracy:** Overall classification accuracy across all classes.
- **Balanced Accuracy:** Given the class imbalance in the dataset, balanced accuracy was a crucial metric to account for performance on underrepresented classes.
- **F1 Score:** Macro F1 scores were computed to evaluate the model's ability to balance precision and recall across all classes.
- **Confusion Matrices:** Used to visualize misclassifications and identify which car makes and models were most challenging for the network. However, due to the significant class imbalance, the confusion matrices may not fully represent the model's performance across all classes, particularly the minority classes. To provide additional context, annotations indicating the number of samples for each class have been included in the confusion matrix plots.
- **ROC Curves:** Micro-averaged ROC curves were generated to assess the overall discrimination ability of the model.

Plots of loss and accuracy curves during training and validation were also generated for all tasks to monitor convergence and diagnose potential overfitting.

VI. RESULTS

In this section, we present the most relevant metrics for car make and car make-model classification. Additional plots and results for each case can be found in the Appendix A and in corresponding GitHub repository.

A. Car Make Classification

A comparison of the results obtained using **CrossEntropy loss** and the best configuration of **Focal loss** (with gamma=1.5, eta=0.4) is shown. Figure 2 displays the **sorted precision, recall, and F1 score per class**. As expected, the Focal loss outperforms CrossEntropy for underrepresented classes, but shows reduced performance for the majority classes.

This behavior is also reflected in Table I, where the **lowest F1 score** is significantly higher in the case of Focal loss. However, the **mean F1 score** is lower compared to CrossEntropy loss. In addition, also accuracy and balanced accuracy are reported, illustrating the trade-off between handling minority and majority classes.



(a) CrossEntropy Loss Results



(b) Focal Loss Results (gamma=1.5, eta=0.4)

Fig. 2: Comparison of F1 Score Results using CrossEntropy and Focal Loss for Make classification.

Statistic	CrossEntropy Loss	Focal Loss (gamma=1.5, eta=0.4)
Accuracy	0.9264	0.8924
Balanced Accuracy	0.8652	0.8265
Highest F1 Score	0.9882	0.9882
Lowest F1 Score	0.3529	0.5000
Mean F1 Score	0.8817	0.8457

TABLE I: Comparison of F1 Scores for CrossEntropy Loss and Focal Loss for Make classification.

B. Fine-Grained Classification

In this section, we present a comparison of results for two different setups:

Case 1: We used a model pretrained on car make classification using CrossEntropy loss. During the fine-grained classification, CrossEntropy loss was applied for both make classification and model classification tasks.

Case 2: We used a model pretrained on car make classification using the best configuration of Focal loss (with gamma=1.5 and eta=0.4). In this case, Focal loss was used for the make classification, while CrossEntropy loss was used for the model classification task.

Table II summarizes the **accuracy results** for both cases across various metrics, it is especially notable that the minimum balanced accuracy for Focal loss is much higher (0.8458 vs. 0.6983). This suggests, again, that Focal loss provides better performance on the most difficult classes, helping to reduce errors on underrepresented categories. On the other hand, CrossEntropy loss generally performs better on the majority classes and overall make accuracy.

Statistic	CrossEntropy Loss	Focal Loss (gamma=1.5, eta=0.4)
Make Accuracy (Not Balanced)	0.9482	0.9382
Make Accuracy (Balanced)	0.9177	0.9036
Model Accuracy (Not Balanced)	0.9346	0.9400
Top-3 Model Accuracy	0.9922	0.9922
Mean Balanced Model Accuracy	0.9352	0.9435
Min Balanced Model Accuracy	0.6983	0.8458
Max Balanced Model Accuracy	1.0000	1.0000

TABLE II: Comparison of Classification Results between Case 1 and Case 2

Exterior Parts				
Part Type	Make Accuracy (Not Balanced)	Make Accuracy (Balanced)	Model Accuracy (Not Balanced)	Top-3 Model Accuracy
Headlight	0.5811	0.4460	0.6305	0.8350
Taillight	0.6280	0.4912	0.8098	0.9371
Fog Light	0.3836	0.2325	0.5851	0.8321
Air Intake	0.8272	0.6211	0.5864	0.8017
Interior Parts				
Part Type	Make Accuracy (Not Balanced)	Make Accuracy (Balanced)	Model Accuracy (Not Balanced)	Top-3 Model Accuracy
Console	0.5867	0.4581	0.6656	0.8696
Steering Wheel	0.1870	0.1692	0.5822	0.8545
Dashboard	0.1545	0.1357	0.5920	0.8736
Gear Lever	0.4374	0.2602	0.5248	0.7683
Weighted Voting Strategy				
Make Accuracy	0.9281			
Make Balanced Accuracy	0.8073			
Model Accuracy	0.7198			
Top-3 Model Accuracy	0.8689			
Mean Balanced Accuracy (Models)	0.7368			

TABLE III: Results for Model Trained Using Different Car Part Type Images and Weighted Voting Strategy

C. Car parts classification

This section presents the results for fine-grained classification using images of different car parts. The accuracy metrics are reported in Table III.

One notable result is the strong performance for the air intake, which may be attributed to the presence of car logos in many of these images, offering a visual clue that aids the classification process. Taillight also performs relatively well, particularly in model accuracy (0.8098), which suggests the model's ability to correctly identify specific car models once the make is correctly classified. In general, exterior parts tend to perform better in terms of make and model accuracy, possibly due to more distinctive external features, while interior parts exhibit more challenging classification scenarios. The results emphasize the importance of visual cues, such as logos, that can significantly aid in improving classification performance. The table also presents the performance of the model when a weighted voting strategy is applied. This strategy combines the results from multiple classifiers, each weighted based on their mean balanced model accuracy. Overall, the weighted voting strategy demonstrates significant improvements in classification accuracy.

VII. CONCLUSIONS

In this work, we have addressed the challenges of fine-grained car classification, particularly focusing on overcoming the limitations posed by class imbalance in large-scale datasets like CompCars. By introducing Focal Loss as an enhancement to the traditional CrossEntropy Loss, we significantly improved classification performance on underrepresented classes, as demonstrated by the increase in the lowest F1 score from 0.3529 (CrossEntropy) to 0.5000 (Focal Loss). Our application of Focal Loss effectively prioritizes hard-to-classify samples, thereby reducing the negative impact of class imbalance, especially in car make classification tasks.

Additionally, we extended the classification task by incorporating car part images (e.g., headlights, taillights, dashboards) to complement full-car image classification. Through the implementation of a weighted voting strategy, which combines the outputs of part-specific models, we achieved an overall

make accuracy of 92.81% and a balanced accuracy of 80.73%, further demonstrating the efficacy of leveraging part-based features in fine-grained classification. The proposed approach has practical implications in real-world car verification systems and car manufacturing processes, where fine-grained classification and addressing class imbalance are crucial. Despite these advances, some challenges remain, particularly in further optimizing the balance between minority and majority class performance. Future work could explore additional strategies for enhancing part-specific models, including more sophisticated data augmentation techniques, or investigating other loss functions that balance minority class emphasis without reducing overall performance. Additionally, expanding the approach to real-time or multi-view car classification could offer further improvements in practical deployment.

The scripts, results, and additional plots generated during this study are available at our GitHub repository, providing further insights into the methodology and enabling reproducibility of the results. .

REFERENCES

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APPENDIX A ADDITIONAL METRIC RESULTS

In this appendix, we present additional metric plots for fine-grained classification using the best configuration of Focal Loss. Similar plots, including those for other configurations and for car parts classification, can be found in the accompanying Github repository.

A. Confusion Matrix Analysis

To further analyze the performance of our model on car make and model classification tasks, confusion matrices were generated for both the **best-performing** and **worst-performing** car makes. The selection of makes and models is based on the following criteria:

- **Make Selection:** Only car makes with multiple models were considered for model classification. The makes with the top 3 highest F1 scores were selected as the best-performing, and the 3 makes with the lowest F1 scores were selected as the worst-performing.
- **Model Selection:** Within each selected make, we ranked models by their classification accuracy, selecting the top 3 best models and the bottom 3 worst models for further analysis.
- **Confusion Matrix Generation:** Confusion matrices were generated for both the best- and worst-performing makes, using true and predicted model labels. These matrices were normalized to display the distribution of correct and incorrect predictions across the selected models.

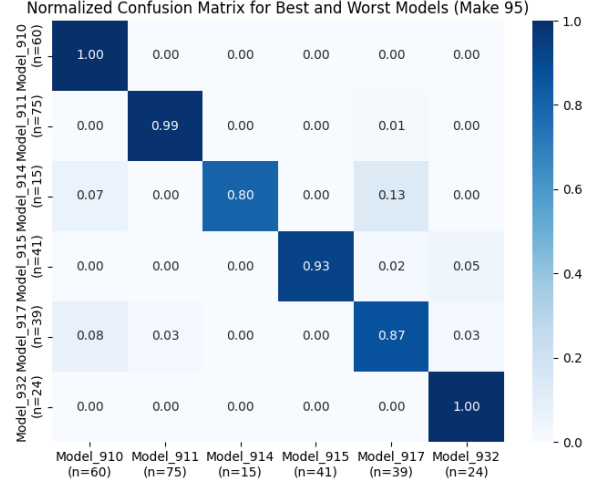
In Figure 3, we present confusion matrices for one of the best and one of the worst-performing car makes. These matrices highlight classification accuracy variations across different models. The class imbalance is annotated by the number of samples for each model, but it is important to note that confusion matrices can sometimes over-represent accuracy for majority classes, making it difficult to fully assess model performance on minority classes.

As seen in Figure 3, the confusion matrix for the best-performing car make shows most predictions falling on the diagonal, indicating high accuracy across selected models. In contrast, the confusion matrix for the worst-performing car make displays more off-diagonal values, reflecting higher levels of misclassification.

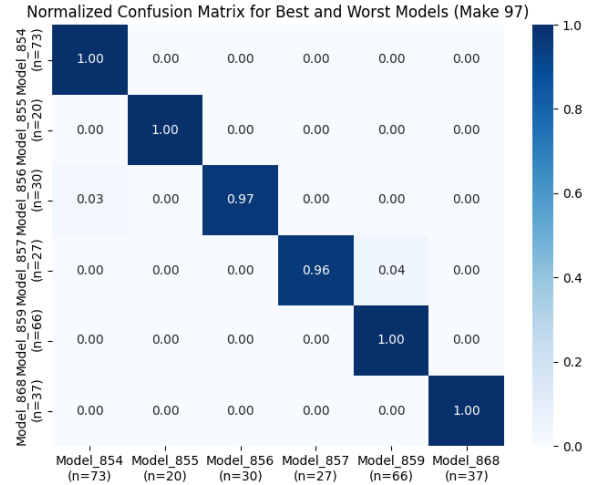
B. Micro-Averaged ROC Curve

In addition to the confusion matrices, **micro-averaged Receiver Operating Characteristic (ROC) curves** were generated for the same car makes. These curves offer a broader perspective on the model’s ability to distinguish between models within each make, particularly when addressing class imbalance. The micro-averaged ROC curve aggregates performance across all classes, providing a single representation of the model’s classification capability.

Figure 4 shows the ROC curves for both the best- and worst-performing car makes. These curves complement the confusion matrix analysis by illustrating the trade-off between true positive rate (sensitivity) and false positive rate. A higher area



(a) Worst-Performing Car Make



(b) Best-Performing Car Make

Fig. 3: Confusion Matrices for Best- and Worst-Performing Car Makes

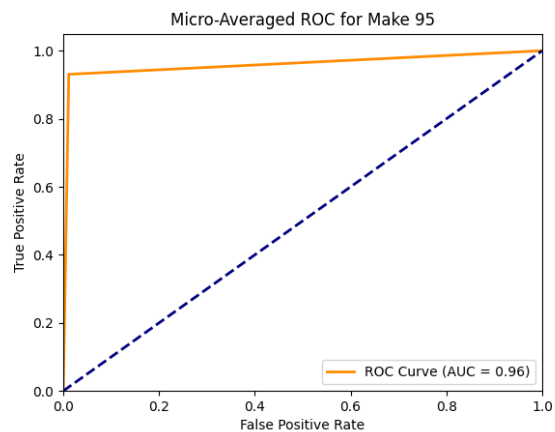
under the curve (AUC) indicates better overall classification performance.

The micro-averaged ROC curve for the best-performing car make shows strong classification ability across all classes, with a higher AUC compared to the worst-performing car make.

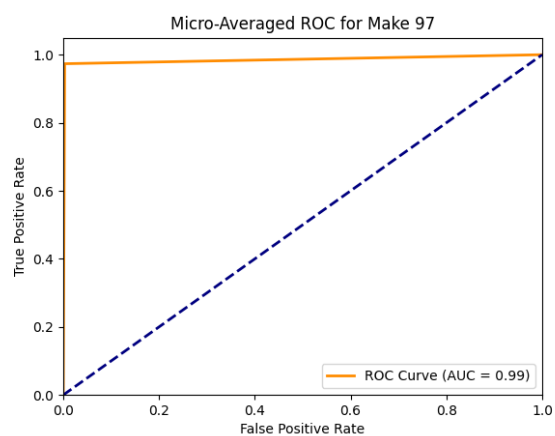
C. Training Convergence Plots

Figure 5 presents the training and validation loss, as well as the accuracy over epochs, for the fine-grained training. These plots demonstrate the model’s convergence behavior, showing its ability to optimize during training and generalize effectively to the validation set.

The training convergence plots provide insight into the learning process of the model, showing how well it adapts during training and validation.



(a) Worst-Performing Car Make



(b) Best-Performing Car Make

Fig. 4: Micro-Averaged ROC Curves for Best- and Worst-Performing Car Makes

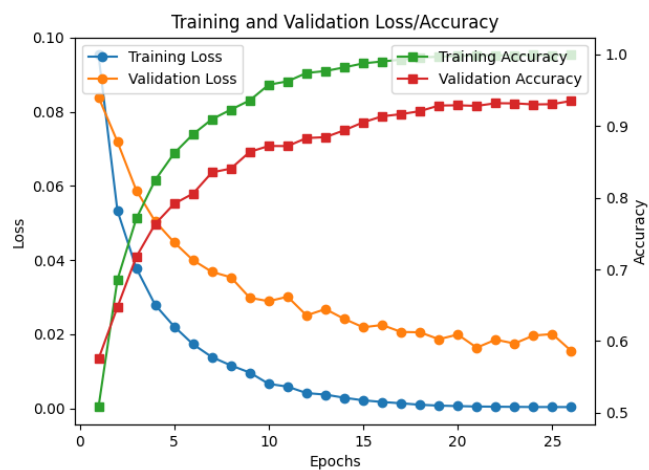


Fig. 5: Training and Validation Loss and Accuracy over Epochs